INTERNSHIP PROJECTREPORT

ELK Stack Based Security Information and Event Management (SIEM) Dashboard

In Partial Fulfilment of the requirements for the Degree of

Master of Computer Applications

In

Cyber Security & Forensics

By

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June 2026

Certificate

Declaration

I, Gurpreet Singh, declare that this project is my original work and has not been submitted

previously for any degree or diploma. The work has been completed under academic guidance and

all referenced sources have been acknowledged.

Student Name: Gurpreet Singh

Signature: ______________________

Date: ___________________________

Acknowledgement

I sincerely thank the faculty of the School of Engineering & Technology, CT University, for their

guidance and encouragement. I also thank my family and friends for their continuous support

throughout the project.

Abstract

This project demonstrates the implementation of a Security Information and Event Management

(SIEM) platform using the ELK Stack (Elasticsearch, Logstash and Kibana). Docker Compose is

used to deploy the complete environment. Logstash ingests and parses sample log files,

Elasticsearch indexes the data, and Kibana provides interactive dashboards for monitoring web

traffic and security events.

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Company Profile

Allsoft Solutions is the organization where this internship was undertaken. It is a technology training and solutions company that works at the intersection of emerging technologies and industry-oriented skill development. The organization is primarily engaged in three major domains: Artificial Intelligence and Machine Learning, Digital Marketing, and Cybersecurity, offering both consulting-oriented services and structured, hands-on training programs in each of these areas. In the Artificial Intelligence and Machine Learning domain, Allsoft Solutions focuses on building practical understanding of data-driven technologies, covering areas such as machine learning fundamentals, model building, and real-world AI applications. The Digital Marketing vertical of the company equips learners with industry-relevant skills in search engine optimization, social media marketing, content strategy, and analytics-driven campaign management, helping individuals and businesses strengthen their online presence. The Cybersecurity division, under which this internship was carried out, focuses on ethical hacking, vulnerability assessment, penetration testing, and network security. This division provides trainees with structured, lab-based exposure to real-world attack and defense techniques using industry-standard tools and frameworks such as the OWASP Testing Guide, Kali Linux, Burp Suite, Nmap, and Metasploit, closely mirroring practices followed in the cybersecurity industry. Akey strength of Allsoft Solutions is its team of experienced and knowledgeable staff, who bring strong subject-matter expertise along with genuine dedication to mentoring interns and trainees. The mentors are approachable, provide consistent guidance, and encourage a hands-on, project-based learning culture rather than a purely theoretical one. This supportive environment made it possible to gain practical exposure to professional cybersecurity workflows, ask questions freely, and receive constructive feedback throughout the course of the internship. Overall, Allsoft Solutions positions itself as a skill-development-focused organization that bridges the gap between academic learning and industry requirements, particularly in fast-evolving technology fields such as AI/ML, digital marketing, and cybersecurity. Being placed in the Cybersecurity team for this internship provided 1direct, practical experience aligned with current industry practices in network traffic analysis and monitoring, which forms the core subject of this report.

Chapter 1: Introduction

Cyber Security focuses on protecting information systems, networks, and applications from

unauthorized access and cyber threats. Modern organizations generate large volumes of log data

from web servers, operating systems, applications, and security devices. A SIEM solution

centralizes these logs, enabling efficient monitoring, investigation, and analysis.

What is SIEM?

Security Information and Event Management (SIEM) combines Security Information Management

(SIM) and Security Event Management (SEM). It collects, normalizes, stores, and analyzes security

logs from multiple sources. SIEM solutions provide centralized visibility, support incident

investigation, and help identify suspicious activity.

Project Objectives

- Implement a basic SIEM platform using the ELK Stack.
- Collect and process web server logs.
- Parse logs using Logstash Grok filters.
- Store events in Elasticsearch.
- Create Kibana dashboards for visualization.
- Demonstrate centralized log monitoring using Docker Compose.

Chapter 2: ELK Stack Overview

The ELK Stack consists of Elasticsearch, Logstash and Kibana. Elasticsearch is a distributed

search and analytics engine that stores indexed log data. Logstash is a data processing pipeline

that collects, parses and forwards logs. Kibana is a visualization platform used to create

dashboards and analyze security events. Docker Compose was used to deploy these services

together, making installation and management simpler.

2.1 Elasticsearch

Elasticsearch stores structured log data as documents in indices. It provides full-text search, fast

queries and aggregations. In this project, processed logs are stored in the 'web-logs' index and

queried by Kibana for visualization.

2.2 Logstash

Logstash reads sample web server logs, applies Grok filters to extract fields such as client IP,

HTTP method, URL, status code and timestamp, and forwards the structured events to

Elasticsearch.

2.3 Kibana

Kibana provides dashboards, Discover, Lens and Visualize tools. The project dashboard displays

Requests Over Time, HTTP Status Codes, Top Client IPs, HTTP Methods and Most Requested

URLs.

Chapter 3: Literature Review

SIEM platforms are widely used in Security Operations Centers (SOCs) to centralize log collection

and incident investigation. The ELK Stack is a popular open-source solution because it is scalable,

flexible and supports powerful search and visualization capabilities. Docker enables rapid

deployment of ELK services across development environments.

Chapter 4: Hardware and Software Requirements

Hardware Requirements:

- Intel Core i5 or higher
- Minimum 8 GB RAM (16 GB recommended)
- 20 GB free disk space

Software Requirements:

- Windows 11
- Docker Desktop
- Visual Studio Code
- Elasticsearch 8.15
- Logstash 8.15
- Kibana 8.15

Chapter 5: System Architecture

Architecture Workflow:

Sample Logs ® Logstash ® Elasticsearch ® Kibana Dashboard

Logstash ingests and parses logs, Elasticsearch indexes the structured events, and Kibana

visualizes the data using interactive dashboards.

Project Workflow

1. Start Docker Desktop.
2. Run **docker compose up -d**.
3. Logstash reads sample log files.
4. Elasticsearch indexes parsed events.
5. Kibana displays dashboards and Discover views.
6. Analysts monitor activity and investigate logs.

Suggested Diagrams for Final Report

- ELK Stack Architecture Diagram
- Docker Container Diagram
- Deployment Diagram
- Data Flow Diagram (Level 0 & 1)
- Use Case Diagram
- Activity Diagram
- Sequence Diagram
- Component Diagram
- Project Folder Structure Diagram

Chapter 6: Docker Installation

Docker Desktop was installed on Windows 11 to run Elasticsearch, Logstash and Kibana inside

isolated containers. Docker Compose was used to start all services using a single configuration file.

Chapter 7: Visual Studio Code Setup

Visual Studio Code was used to create and edit the project files including docker-compose.yml,

Logstash pipeline configuration and sample log files. The integrated terminal was used to execute

Docker commands.

Project Folder Structure

project/

nnn docker-compose.yml

nnn logstash/

n nnn pipeline/

n nnn logstash.conf

nnn sample_logs/

n nnn access.log

n nnn auth.log

docker-compose.yml

The docker-compose.yml file defines three services: Elasticsearch – stores indexed logs. Logstash

– parses and forwards logs. Kibana – visualizes indexed data. A Docker volume is used to persist

Elasticsearch data between restarts.

Logstash Pipeline (logstash.conf)

The Logstash pipeline contains three stages: **Input**: Reads log files from the mounted sample_logs

directory. **Filter**: Uses Grok patterns to extract fields such as client_ip, request, method, status and bytes. **Output**: Sends parsed events to Elasticsearch index **web-logs** and prints them to the

console for debugging.

Elasticsearch Configuration

Elasticsearch runs in single-node mode with security disabled for development. It indexes

processed log events and provides REST APIs for searching and aggregating data.

Kibana Configuration

Kibana connects to Elasticsearch using the ELASTICSEARCH_HOSTS environment variable. A

data view named **web-logs** was created to explore indexed documents and build visualizations.

Project Execution Commands

Start the ELK Stack

docker compose up -d

Check running containers

docker ps

Restart Logstash

docker compose restart logstash

Stop containers

docker compose down

Expected Output

After starting the stack: Elasticsearch is available at http://localhost:9200 Kibana is available at

http://localhost:5601 Logstash ingests sample logs into the **web-logs** index. Kibana Discover

displays indexed documents.

Figure 4.1: Starting ELK Stack using Docker

Compose.

This screenshot shows the Docker Compose command being executed to start Elasticsearch,

Logstash, and Kibana containers.

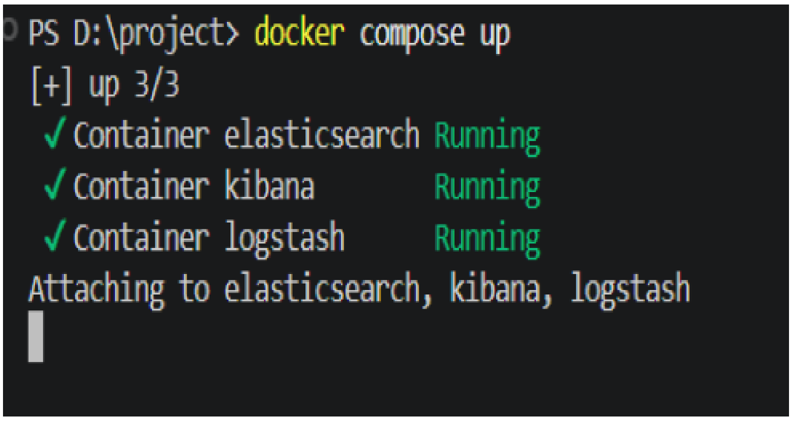

Figure 4.2: Docker Compose Execution.

The services are launched in detached mode. Docker automatically creates the required network

and starts each service.

Figure 4.3: Elasticsearch Verification.

Opening http://localhost:9200 confirms Elasticsearch is running and responding to REST API

requests.

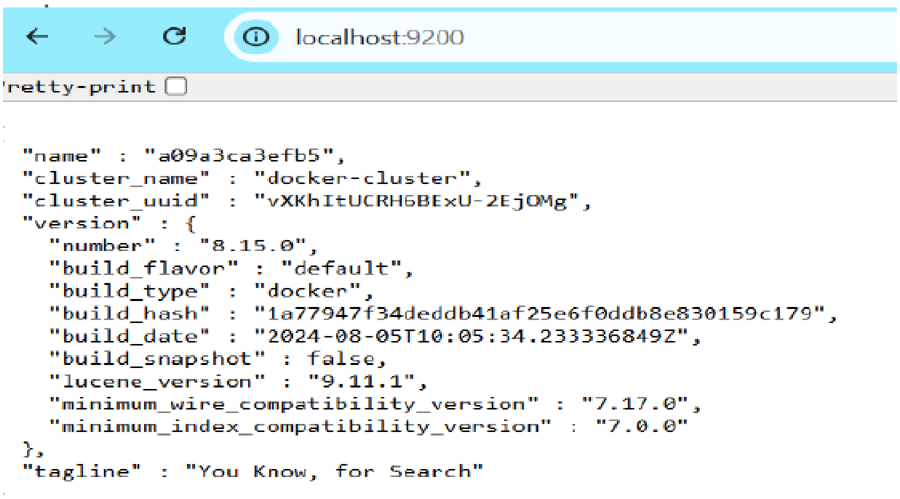

Figure 4.4: Kibana Home Page.

Kibana provides the graphical interface for searching data, creating dashboards, and visualizing

events.

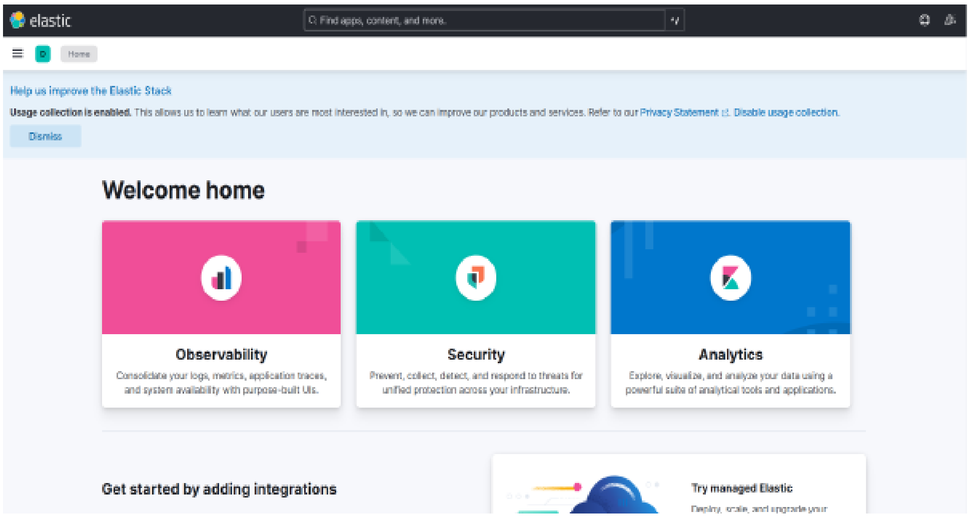

Figure 4.5: Indexed Documents.

The Discover page displays documents stored in the web-logs index after processing by Logstash.

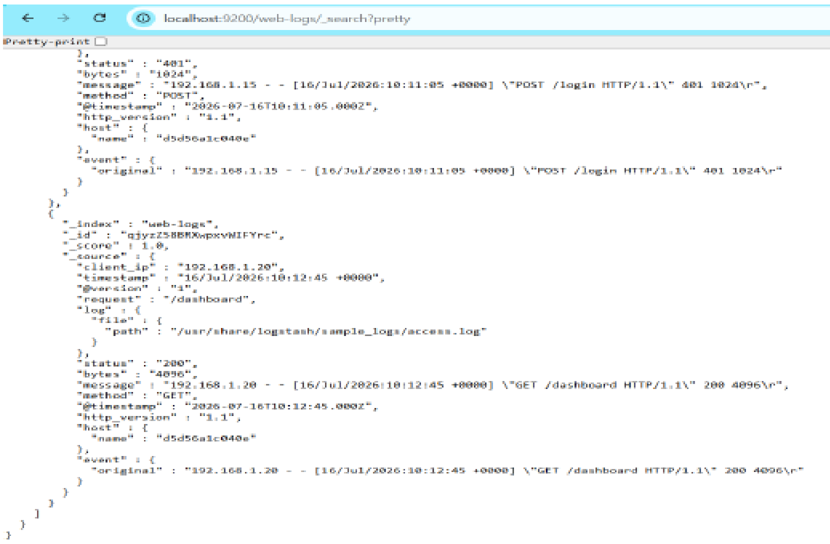

Figure 4.6: SIEM Dashboard.

The dashboard combines multiple visualizations into a single monitoring interface.

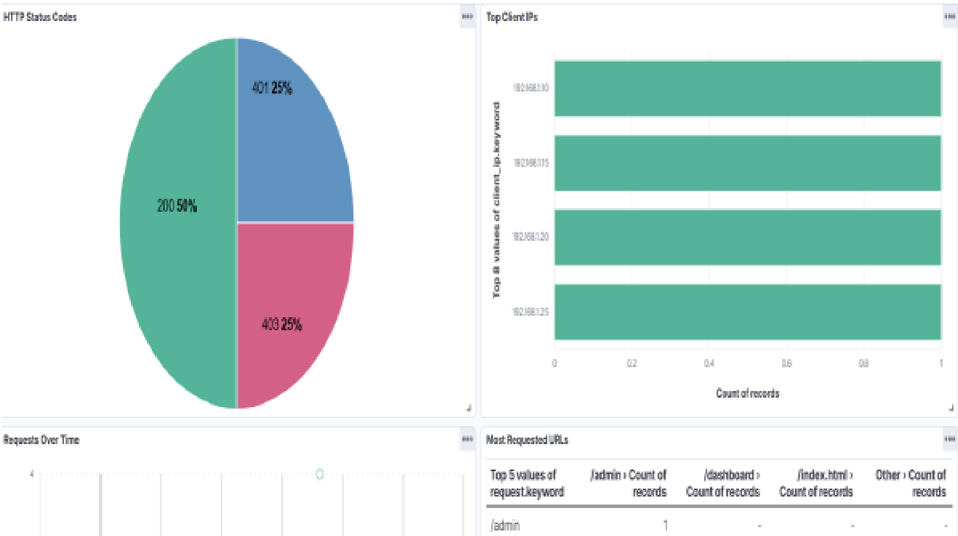

Figure 4.7: HTTP Methods Visualization.

This visualization summarizes GET and POST requests, helping analysts understand request

patterns.

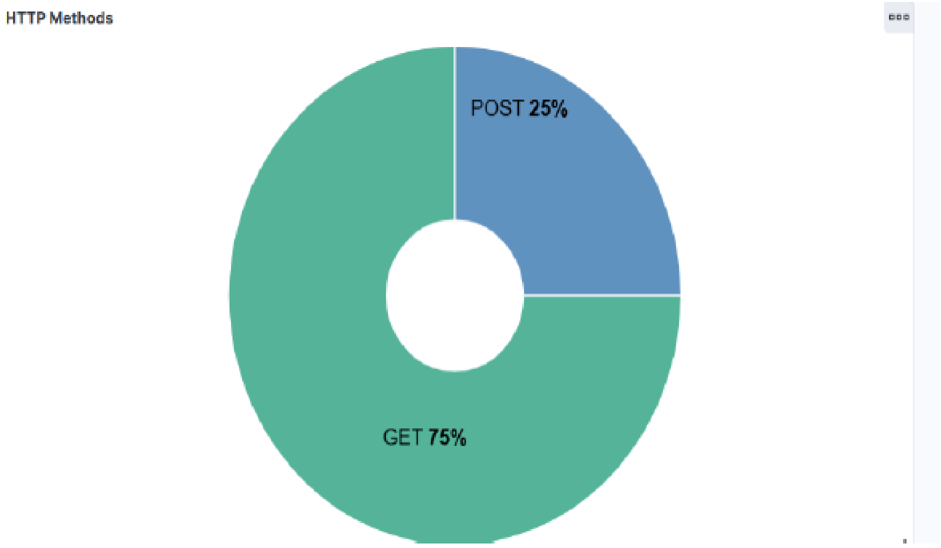

Figure 4.8: Requests Over Time.

A time-series chart displays how requests change over time, useful for identifying spikes in activity.

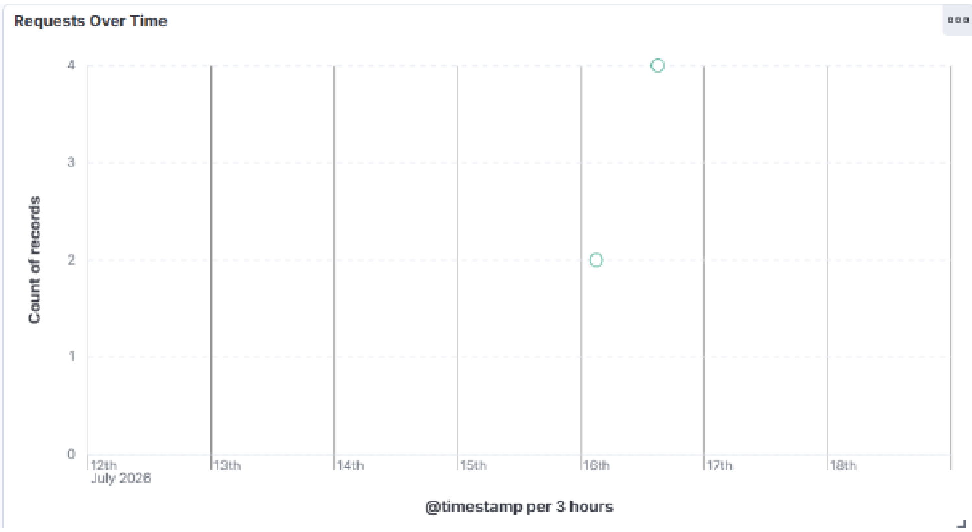

Figure 4.9: Most Requested URLs.

A table lists the most frequently requested resources, helping identify popular or suspicious

endpoints.

Most Requested URLs

Top 5 values of request.keyword	/admin › Count of records	/dashboard › Count of records	/index.html › Count of records	Other › Count of records
/admin	1	-	-	-
/dashboard	-	1	-	-
/index.html	-	-	1	-
/login	-	-	-	1

Dashboard Analysis

The completed SIEM dashboard provides centralized monitoring of web logs. The HTTP Status

Code chart highlights successful and failed requests, the Top Client IP chart identifies the most

active systems, Requests Over Time visualizes traffic trends, HTTP Methods separates GET and

POST activity, and the Most Requested URLs table identifies frequently accessed resources.

Together these visualizations demonstrate how the ELK Stack can be used to collect, process,

index, and analyze security-relevant log data in a single interface.

ELK Stack Based Security Information and Event

Management (SIEM) Dashboard

References

Chapter 8: Testing and Results

The project was tested after deploying Elasticsearch, Logstash and Kibana using Docker Compose.

Sample web server logs were successfully parsed by Logstash and indexed into Elasticsearch.

Kibana Discover displayed the indexed documents and the custom dashboard visualized the

collected data. The tests confirmed that the ELK stack was functioning correctly and that logs could

be searched and analyzed efficiently.

Functional Test Cases

Test 1: Start ELK Stack using **docker compose up -d** – **Result:** Passed

Test 2: Verify running containers using **docker ps** – **Result:** Passed

Test 3: Access Elasticsearch on **http://localhost:9200** – **Result:** Passed

Test 4: Access Kibana on **http://localhost:5601** – **Result:** Passed

Test 5: Verify indexed logs in Discover – **Result:** Passed

Test 6: Verify dashboard visualizations – **Result:** Passed

Advantages

• Centralized log collection and monitoring.

• Fast indexing and search using Elasticsearch.

• Interactive dashboards in Kibana.

• Open-source and cost-effective solution.

• Easy deployment with Docker Compose.

• Suitable for cybersecurity learning and demonstrations.

Limitations

• Uses sample log data instead of production data.

• Basic Logstash parsing rules only.

• No automated alerting configured.

• No machine learning or threat intelligence integration.

• Designed primarily for educational purposes.

Future Scope

Future improvements include integrating Filebeat, Winlogbeat, Suricata IDS, firewall logs, email

alerting, Elastic Security, anomaly detection, machine learning, and threat intelligence feeds to build

a more advanced enterprise SIEM solution.

Conclusion

The ELK Stack Based SIEM Dashboard successfully demonstrates how security logs can be

collected, parsed, indexed and visualized using open-source technologies. Docker Compose

simplified deployment, Logstash processed the log files, Elasticsearch stored and searched the

events, and Kibana provided meaningful dashboards. The project meets its objectives and provides

practical knowledge of SIEM implementation for MCA Cyber Security students.

References

1. Elasticsearch Official Documentation.

2. Logstash Official Documentation.

3. Kibana Official Documentation.

4. Docker Official Documentation.

5. Elastic Stack Documentation.

6. OWASP Logging Cheat Sheet.

7. NIST Cybersecurity Framework.

Appendix – Useful Commands

Start Project

docker compose up -d

Stop Project

docker compose down

Restart Logstash

docker compose restart logstash

View Running Containers

docker ps

View Logstash Logs

docker logs logstash

Open Elasticsearch

http://localhost:9200

Open Kibana

http://localhost:5601